The Cost of Forgetting

Counterfactual Realization Entropy, Slow Memory,
and Open Problems in Substrate Skepticism

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Priority convention. This note makes no claim of priority. Quite the opposite: as a working rule, every mathematically correct statement below should be presumed to have been proved before, possibly in a different language, until a serious literature search establishes otherwise. Several ingredients are certainly classical or already explicit in adjacent literatures. If a precise antecedent is found for one of the statements isolated here, the appropriate response is to cite it and reinterpret the present note as an independent translation into the response-tree/minimum-entropy-coupling framework.

Abstract

We study a narrow question that survives after stripping away several claims that are already covered by minimum-entropy coupling, random-map realizations of Markov chains, causal and bicausal coupling, stationary joinings, counterfactual simulation, and long-memory process theory. For a finite family of policies, let κ_n denote the minimum Shannon entropy of an arbitrary common source capable of reproducing all policy transcript laws, and let χ_n denote the minimum entropy of a single source that instantiates one globally coherent deterministic response tree. Their difference

$$\Gamma_n := \chi_n - \kappa_n$$

is the *counterfactual coherence premium* relative to the unconstrained coupling benchmark.

The main structural observation is that the asymptotics of Γ_n appear to be controlled not by "ergodicity" alone, but by how rapidly the process forgets finite boundary information in an exact coupling sense. Primitive finite-state controlled Markov systems admit a natural bounded-premium argument, whereas stationary ergodic renewal-reward constructions with heavy-tailed durations give unbounded lower bounds as large as $n/L(n)$ for slowly varying $L \to \infty$. This leaves a sharper open problem: whether stationary ergodicity alone forces $\Gamma_n = o(n)$, and, more generally, which sublinear growth laws are realizable.

The philosophical consequence is deliberately modest. No inference to fundamentality follows. What survives is an interface-relative version of substrate skepticism: once the realization contract is fixed, one may ask what information any exact substrate must supply, retain, or encode. The relevant object may be not a single entropy rate but an entire *forgetting profile* $n \mapsto \Gamma_n$.

1 What is deliberately not claimed

The conceptual neighborhood is heavily occupied. Minimum-entropy coupling and coupling-based entropy pseudometrics are established objects [1]. Asymptotic coupling problems, including minimum-entropy coupling for product sources, have been studied in depth [2]. Modern approximation results introduce profile-based lower bounds and constant-additive approximations for finite MEC instances [3]. Random-map realizations of finite ergodic Markov chains, including an explicit notion of entropy redundancy, are already present in Yano and

Yasutomi [4]. Causal or bicausal restrictions on process couplings are also established [5], as are stationary optimal couplings (joinings) for stochastic processes [6].

In the controlled/counterfactual direction, recent work makes explicit that ordinary MDP marginals do not determine a joint law of counterfactual outcomes across actions [9]. Closely related work on common random numbers shows that sharing a pseudorandom seed across intervention scenarios can still be causally incoherent when draw identities depend on execution path, and repairs this by event-keyed randomness [10].

Accordingly, this note does *not* claim as new any of the following ideas:

- minimum-entropy coupling;
- entropy as a cost of stochastic realization;
- random-map realization of Markov dynamics;
- the distinction between marginal correctness and cross-world/counterfactual coherence;
- causal or adapted coupling constraints;
- long-memory renewal constructions;
- stationary joinings or entropy-rate methods.

The only purpose of the present framework is to isolate a specific difference of two entropy minima and ask how its growth is controlled by dynamical forgetting.

2 Finite-horizon accounting

Let W be a finite controlled stochastic process, G a finite family of deterministic policies, and T_n^π the length-n transcript generated under policy $\pi \in G$, with law μ_n^π.

Definition 2.1 (Outcome coupling cost)**.** *Define*

$$\kappa_n^G(W) := \min_{P \in \mathcal{C}(\mu_n^\pi : \pi \in G)} H_P\big((X_\pi)_{\pi \in G}\big),$$

where $\mathcal{C}$ denotes the set of couplings with the prescribed policy marginals.

Equivalently, κ_n^G is the minimum entropy of a common source E from which each policy transcript can be produced by its own deterministic decoder. The decoders need not interpret a source atom consistently across policies.

Definition 2.2 (Response-tree realization cost)**.** *Let $TC_n(G)$ denote the set of policy tuples compatible with one deterministic causal response tree through horizon n. Define*

$$\chi_n^G(W) := \min_{\substack{P \in \mathcal{C}(\mu_n^\pi : \pi \in G) \\ \operatorname{supp} P \subseteq TC_n(G)}} H(P).$$

The counterfactual coherence premium *is*

$$\Gamma_n^G(W) := \chi_n^G(W) - \kappa_n^G(W) \geq 0.$$

The distinction is operational. κ_n asks for one compressed source serving many policy marginals. χ_n additionally requires the source value to specify one coherent world in which equal history–action nodes receive equal responses.

2.1 No re-merging of full histories

The full-history convention has a simple but decisive consequence.

Lemma 2.3 (No re-merging)**.** *If two deterministic policy paths choose different actions at some time τ, then their full histories are unequal at every later time:*

$$A_\tau^\pi \neq A_\tau^\sigma \quad \Longrightarrow \quad h_t^\pi \neq h_t^\sigma \quad \forall t > \tau.$$

Proof. The action at time τ is part of every subsequent full history. Hence the two histories cannot again be identical as finite strings. □

Thus cross-policy response-tree constraints can only occur before the policies have separated in full history. This elementary fact drives both the bounded-premium Markov argument and the slow-forgetting counterexamples below.

3 A bounded-premium regime: primitive finite-state dynamics

This section records a candidate theorem that appears to follow from standard finite-state coupling machinery together with the no-remerging lemma. No priority claim is made.

Assume that W is a fully observed finite-state time-homogeneous controlled Markov system with transition kernel

$$P(s' \mid s, a),$$

and that G is a fixed finite family of stationary deterministic policies $\pi : S \to A$. Let

$$P_\pi(s, s') := P(s' \mid s, \pi(s)).$$

Proposition 3.1 (Bounded premium under primitive policy chains — candidate statement)**.** *Suppose every P_π, $\pi \in G$, is irreducible and aperiodic. Then there should exist a finite constant $C(W, G)$ such that*

$$0 \leq \Gamma_n^G(W) \leq C(W, G) \qquad \textit{for all } n.$$

Consequently,

$$\frac{\Gamma_n^G(W)}{n} \longrightarrow 0.$$

Proof mechanism. For distinct policies define their disagreement set

$$D_{\pi\sigma} := \{s : \pi(s) \neq \sigma(s)\}.$$

Until the common path first visits $D_{\pi\sigma}$, the two policies choose the same action and therefore evolve on the same response-tree branch. For a finite irreducible aperiodic chain, hitting a nonempty set has a geometrically decaying tail uniformly over starting states. Hence the pairwise split time $\tau_{\pi\sigma}$ has finite expectation and finite entropy. Since G is finite, the maximum split time

$$T := \max_{\pi \neq \sigma} \tau_{\pi\sigma}$$

also has finite expectation and entropy.

Let B be the random finite response-tree skeleton generated up to T. Because at most $|G|$ active policy paths are present at each time,

$$H(B) \leq H(T) + C_0 \, \mathbb{E}[T + 1] < \infty$$

for a finite alphabet-dependent constant C_0. After T, all distinct policy histories live in disjoint full-history subtrees, so no further cross-policy equality constraints remain.

The second ingredient is forgetting of initialization. Standard coupling of finite primitive Markov chains gives, for each π, a finite constant K_π such that the m-block law from any starting state and the stationary m-block law are at uniformly bounded coupling-entropy distance, independent of m. The corresponding multi-marginal MEC values therefore differ by at most $\sum_\pi K_\pi$. Combining this with the finite entropy of B yields a bound of the schematic form

$$\Gamma_n^G \leq H(B) + 2 \sum_{\pi \in G} K_\pi.$$

The important point is not the precise constant. It is that the full-history coherence constraint is confined to a finite-information transient when the induced policy dynamics forget their initial conditions at a geometric rate.

Remark 3.2. *This proposition must not be confused with the statement that an ergodic Markov realization has zero or bounded* total *realization redundancy. Random-map realization can itself have a positive entropy-rate overhead [4]. The bounded quantity here is only the extra surcharge*

$$\chi_n - \kappa_n$$

created by requiring one globally coherent response tree rather than an arbitrary common coupling of the same policy marginals.

4 Stationary ergodicity does not imply bounded premium

The previous argument depends on fast forgetting, not on ergodicity by itself. A stationary ergodic process can remember a finite boundary condition for a very long time.

4.1 Heavy-tailed renewal-reward environment

Let $(S_t)_{t\in\mathbb{Z}}$ be a stationary renewal-reward process taking values in $\{-1, +1\}$. Renewal intervals have i.i.d. lengths $T_1, T_2, \ldots$ with

$$\mu := \mathbb{E}[T_1] < \infty,$$

and each renewal block receives an independent fair sign $W_k \in \{-1, +1\}$, constant throughout the block. The equilibrium version is stationary; heavy-tailed duration-driven long-range dependence of precisely this type is classical [7].

At each time the agent observes S_t and chooses an action $A_t \in \{0, 1\}$. The environment then produces an auxiliary output Z_t:

$$A_t = 0 \Rightarrow Z_t \sim \text{Bernoulli}(1/2), \qquad A_t = 1 \Rightarrow Z_t = 0.$$

The fresh fair bits used when $A_t = 0$ are independent of the renewal process.

Consider the crossed stationary policies

$$\pi(+1) = 0, \qquad \pi(-1) = 1,$$

$$\sigma(+1) = 1, \qquad \sigma(-1) = 0.$$

The initial sign S_0 is observed at a common pre-action node. Thus a tree-compatible realization must preserve the same S_0 across both policy coordinates, while an unconstrained MEC may pair atoms across opposite initial signs.

Let α_n denote the conditional policy-tail law under π given $S_0 = +1$, and let β_n denote the corresponding tail law under σ. Sign symmetry swaps these two laws when $S_0 = -1$. The crossed two-context calculation therefore yields

$$\Gamma_n = \text{MEC}(\alpha_n, \beta_n) - \frac{H(\alpha_n) + H(\beta_n)}{2}.$$

Since any coupling has joint entropy at least the larger marginal entropy,

$$\Gamma_n \geq \frac{1}{2} \left| H(\alpha_n) - H(\beta_n) \right|.$$

For a fixed sign path, policy π receives one fresh bit at times with $S_t = +1$, whereas σ receives one fresh bit at times with $S_t = -1$. The entropy of the sign path itself cancels, giving

$$H(\alpha_n) - H(\beta_n) = \sum_{t=0}^{n-1} \mathbb{E}[S_t \mid S_0 = +1].$$

By sign symmetry,

$$\mathbb{E}[S_t \mid S_0 = +1] = \mathbb{E}[S_0 S_t] =: \rho(t).$$

Hence

$$\boxed{\Gamma_n \geq \frac{1}{2} \sum_{t=0}^{n-1} \rho(t).} \tag{1}$$

For the equilibrium renewal-reward process, S_0 and S_t remain correlated exactly when no renewal separates them. Thus, up to the conventional integer indexing,

$$\rho(t) = \frac{1}{\mu} \sum_{\ell > t} \Pr(T_1 \geq \ell).$$

This identifies the coherence-premium lower bound with an integrated residual-life tail.

4.2 Power-law and almost-linear lower bounds

Suppose

$$\Pr(T_1 \geq m) \sim c\, m^{-\alpha}, \qquad 1 < \alpha < 2.$$

Then

$$\rho(t) \sim \frac{c}{\mu(\alpha - 1)} t^{1-\alpha},$$

and therefore

$$\sum_{t=1}^{n} \rho(t) \sim \frac{c}{\mu(\alpha - 1)(2 - \alpha)} n^{2-\alpha}.$$

Equation (1) gives

$$\boxed{\Gamma_n \geq \frac{c}{2\mu(\alpha - 1)(2 - \alpha)}\, n^{2-\alpha}(1 + o(1)).}$$

Thus every polynomial exponent $\beta \in (0, 1)$ occurs as a lower-bound exponent by choosing $\alpha = 2 - \beta$.

Even slower integrable duration tails push the lower bound arbitrarily close to linear. For example,

$$\Pr(T_1 \geq m) \asymp \frac{1}{m(\log m)^2}$$

has finite mean but equilibrium residual tail

$$\rho(t) \asymp \frac{1}{\log t},$$

so

$$\Gamma_n \gtrsim c \frac{n}{\log n}.$$

Iterated logarithms yield still slower forgetting, e.g.

$$\Pr(T_1 \geq m) \asymp \frac{1}{m \log m (\log \log m)^2} \quad \Rightarrow \quad \Gamma_n \gtrsim c \frac{n}{\log \log n}.$$

The lesson is sharp:

$$\boxed{\text{stationary ergodicity alone does not force } \Gamma_n = O(1).}$$

What it appears to control, if anything, is the possibility of a strictly positive *linear* coherence rate.

5 The central unknown: can stationary ergodicity support a positive rate?

Define

$$\gamma(W, G) := \limsup_{n \to \infty} \frac{\Gamma_n^G(W)}{n}.$$

The renewal construction above permits $\Gamma_n \to \infty$ and growth arbitrarily close to linear, while retaining finite alphabets and stationary ergodicity. It does not produce $\gamma > 0$.

Open Problem 5.1 (Stationary-ergodic positive-rate problem)**.** *Does there exist a finite-alphabet stationary ergodic controlled process and a fixed finite policy family G such that*

$$\gamma(W, G) > 0?$$

A natural opposite hypothesis is the following.

Conjecture 5.2 (Zero-rate conjecture)**.** *For a suitable finite-alphabet stationary ergodic response-tree model, with a fixed finite family of deterministic policies that are stationary or shift-covariant,*

$$\Gamma_n^G = o(n).$$

Equivalently,

$$\gamma(W, G) = 0.$$

The conjecture is intentionally stated cautiously because several technical choices matter: how stationary policies are represented on full histories, what behavioral equivalence means, and what exact asymptotic coupling theorem is available for the conditional future laws after finite prefixes.

A naive entropy-rate argument is insufficient. Equal entropy rates do not by themselves make exact finite-block couplings trivial; asymptotic MEC and exact distribution matching have nontrivial information-spectrum structure [2]. Finitary isomorphism results for equal-entropy processes likewise show that exact conversion can carry substantial coding-radius complexity even when entropy rates coincide [8]. Thus any proof of the zero-rate conjecture must control more than Shannon entropy rates.

6 A proposed organizing object: the forgetting profile

The renewal calculation suggests that the key variable is neither state-space size nor ergodicity as such, but the persistence of finite boundary information.

For a process family, let B denote a finite piece of past information that is shared by multiple policy branches before they separate. Define schematically

$$F_B(n) := \max_{b,b'} d_{\mathrm{CE}}\left(\mathrm{Law}(X_{1:n} \mid B = b), \mathrm{Law}(X_{1:n} \mid B = b')\right),$$

where

$$d_{\mathrm{CE}}(P, Q) := \inf_{(X,Y)} [H(X \mid Y) + H(Y \mid X)]$$

is a coupling-based entropy pseudometric of the type studied in [1].

The exact definition should depend on the controlled response-tree model, but the heuristic relation is

$$\Gamma_n \lesssim \underbrace{H(B)}_{\text{shared skeleton}} + \underbrace{\mathcal{F}_n(B)}_{\text{cost of forgetting boundary information}}.$$

This suggests three dynamical regimes:

$$\begin{aligned} F_B(n) &= O(1) && \Rightarrow && \text{bounded coherence premium is plausible,} \\ F_B(n) \to \infty, \quad F_B(n) &= o(n) && \Rightarrow && \text{unbounded but subextensive premium,} \\ F_B(n) &= \Theta(n) && \Rightarrow && \text{positive coherence rate becomes possible.} \end{aligned}$$

The first implication is realized by primitive finite-state chains; the middle regime is realized at the level of lower bounds by heavy-tailed renewal memory; the third occurs in nonergodic persistent-regime constructions.

Open Problem 6.1 (Forgetting criterion)**.** *Find necessary and sufficient conditions, stated in intrinsic process terms, for*

$$\sup_n \Gamma_n < \infty, \qquad \Gamma_n = o(n), \qquad \textit{and} \qquad \limsup_n \Gamma_n/n > 0.$$

Open Problem 6.2 (Growth-spectrum problem)**.** *Which sublinear functions $g(n)$ can occur, up to asymptotic equivalence or constant factors, as*

$$\Gamma_n \asymp g(n)$$

for finite-alphabet stationary ergodic controlled processes? Can every sufficiently regular unbounded $g(n) = o(n)$ be realized?

The renewal family suggests a surprisingly broad spectrum, including powers n^β and almost-linear forms such as $n/\log n$ and $n/\log\log n$. Determining matching upper bounds is a separate problem.

7 Representation dependence and substrate invariance

The phrase "information cost of reality" is meaningless until the realization contract is specified. The same observable process may have very different costs under different allowed implementations. Random-map representations provide a concrete warning: an ergodic Markov chain can require extra entropy when constrained to a particular random-map architecture [4]. Likewise, counterfactual simulators can be marginally correct while differing in whether their shared randomness is attached coherently to modeled events [10].

This motivates an invariance problem.

Open Problem 7.1 (Representation invariance)**.** *Which asymptotic features of*

$$n \longmapsto \Gamma_n$$

are invariant under reasonable recodings of the internal realization, such as finite-memory invertible recodings, equivalent finite-state presentations, or finitary isomorphisms? In particular, under what class of recodings is

$$\gamma = \limsup_n \Gamma_n / n$$

invariant?

If γ is robust under a sufficiently broad class of implementation changes, it becomes a candidate *substrate-skeptical invariant*: a quantity that says something about any exact realization in the class without identifying what the substrate is made of.

If, by contrast, γ can be arbitrarily changed by innocuous recodings, then it is not a property of the world-interface itself but of a chosen bookkeeping convention. This distinction is essential before assigning philosophical weight to any entropy bill.

8 Substrate skepticism after the audit

The framework does not establish

$$\mathrm{Obj}(W) \Rightarrow \mathrm{Fund}(W),$$

nor does it establish that exact simulation is impossible. Observable randomness and causal consistency constrain realizations only relative to an explicit resource contract.

A finite-entropy discrete seed E cannot deterministically generate transcript entropies growing without bound, since

$$H(T_n) \leq H(E).$$

Hence an exact indefinitely stochastic world requires, under that accounting standard, either an unbounded pre-existing information reservoir or continued access to additional information. But this conclusion changes if the substrate is allowed infinite-precision continuous variables: a single $U \sim \mathrm{Unif}[0, 1]$ can encode an infinite binary tape in its expansion. The bill has not disappeared; the unit of account has changed from finite Shannon information to infinite precision.

The conservative principle is therefore:

$$\boxed{\text{The bill is real only after the accounting standard is fixed.}}$$

Once the standard is fixed, the forgetting profile suggests a more refined question than "how many random bits are needed?" A substrate may need to retain information not because the observable process remains macroscopically different forever, but because exact counterfactual coherence preserves distinctions between histories that the observable dynamics itself slowly forgets.

This yields a substrate-skeptical reading of the three regimes:

Growth	Interpretation under the chosen realization contract
$\Gamma_n = O(1)$	Coherence is a finite startup or transient charge.
$\Gamma_n \to \infty$, $\Gamma_n = o(n)$	Coherence has no positive asymptotic rate, but a growing memory debt persists.
$\Gamma_n = \Theta(n)$	Coherence requires an extensive long-run resource.

The philosophical claim is intentionally weaker than an ontology:

> We do not know what lies beneath the world. But once we specify what an exact substrate must reproduce, and which resources it is allowed to use, we can ask how long it must remember distinctions that the observable world appears to forget.

This is the sense in which the problem connects to substrate skepticism. The target is not the identity of the substrate, but invariants of the realization interface.

9 Research program

After removing claims covered by existing literature, the following questions remain the most useful targets.

1. **Stationary-ergodic rate.** Determine whether $\gamma > 0$ is possible in the finite-alphabet stationary ergodic setting, or prove a zero-rate theorem.
2. **Sharp renewal asymptotics.** For the heavy-tailed crossed-policy renewal construction, replace the lower bounds by matching asymptotics for Γ_n. This requires control of the MEC term, not merely the difference of marginal entropies.
3. **Growth classification.** Determine which functions between $O(1)$ and $\Theta(n)$ can occur as coherence-premium growth laws.
4. **Intrinsic forgetting criterion.** Find a process-level condition equivalent to bounded, sublinear, or linear Γ_n, preferably formulated through an established coupling, mixing, coding, or ergodic-theoretic invariant.
5. **Representation robustness.** Determine what part of the growth profile survives changes of finite-state presentation or finitary recoding.
6. **Locate the missing paper.** Assume, until evidence to the contrary, that at least some of the above has already been proved under different terminology. The first genuine research task is therefore bibliographic as well as mathematical: identify the closest exact antecedent and shrink the claim accordingly.

10 Conclusion

The useful distinction is no longer simply between outcome compression and realization compression. The emerging question is dynamical:

> How much information is required to preserve exact coherence while the process forgets its past?

Primitive finite-state mixing suggests a finite transient charge. Heavy-tailed stationary ergodic memory produces an unbounded but potentially sublinear charge. Persistent regime memory permits an extensive charge. The boundary between these cases is the open mathematical object.

No claim is made that this boundary, or even the particular formulations used here, are historically new. The safer presumption is that the correct theorem already exists somewhere. Until its owner is found, the present note records the map of the problem and the exact place where the map still says: *reference missing.*

References

[1] M. Kovačević, I. Stanojević, and V. Šenk, *On the Entropy of Couplings*, arXiv:1303.3235, 2013.

[2] L. Yu and V. Y. F. Tan, *Asymptotic Coupling and Its Applications in Information Theory*, arXiv:1712.06804, 2017.

[3] S. Compton, D. Katz, B. Qi, K. Greenewald, and M. Kocaoglu, *Minimum-Entropy Coupling Approximation Guarantees Beyond the Majorization Barrier*, arXiv:2302.11838, 2023.

[4] K. Yano and K. Yasutomi, *Random walk in a finite directed graph subject to a synchronizing road coloring*, arXiv:1105.1095, 2011.

[5] V. Moulos, *Bicausal Optimal Transport for Markov Chains via Dynamic Programming*, arXiv:2010.06831, 2020.

[6] K. O'Connor, K. McGoff, and A. B. Nobel, *Estimation of Stationary Optimal Transport Plans*, arXiv:2107.11858, 2021.

[7] M.-C. Hsieh, C. M. Hurvich, and P. Soulier, *Asymptotics for Duration-Driven Long Range Dependent Processes*, Journal of Econometrics 141(2), 913–949, 2007; arXiv:0705.2701.

[8] U. Gabor, *Moment-optimal finitary isomorphism for i.i.d. processes of equal entropy*, arXiv:2412.15425, 2024.

[9] E. C. Kaya, M. Ghasemi, and A. Hashemi, *Joint MDPs and Reinforcement Learning in Coupled-Dynamics Environments*, arXiv:2603.06946, 2026.

[10] V. Buffalo, C. A. B. Pearson, and D. Klein, *Realizing Common Random Numbers: Event-Keyed Hashing for Causally Valid Stochastic Models*, arXiv:2603.11084, 2026.